How you'd explain the chain to a person.

The question is going to be: why not a database? If you can tell that story like you're explaining it at a table, you're fine. The fancy names come after the story, not before.

01 You're not protecting a picture. You're protecting a date.

> THE SCENE
> Two years later, a brand gets sued over an image. Their defence is: we checked it on 2 September, it was clear. That defence is a date. The date is sitting on our servers. We are the vendor they paid. If we can quietly move the date, the defence is a story we told about ourselves.

The AI answers "what's true about this picture?" The chain answers a ruder question: what can nobody change about what we already said — including us?

We do not put the picture on a blockchain. That would be insane. We don't put the customer's name up there either. We put a wax seal of the answer. Tiny. Meaningless without the letter. Impossible to fake if you've got the letter in your hand.

02 The school register vs the noticeboard

A database is the school register

We hold the only pen. We can mark someone present. We can also, in October, change it to September and nobody walking in can tell the difference. That's not Postgres being evil. That's what a database is: a notebook with an owner.

The chain is the public noticeboard

Once it's pinned, copies exist all over the place. We don't have a master eraser. An opposing lawyer doesn't have to like us, or log into us. They read the board.

> THE TWO SENTENCES
> A database answers "what do you say happened?" A chain answers "what can you no longer change about what you said?"
>
> Our customer's defence is a date, and we're the ones getting paid to hold it. We need a witness that isn't us.

03 The three follow-ups, answered like a human

"Just make the table append-only."

Cute. We hold the password that makes it append-only, so we also hold the password that turns that off. You've moved the trust into our ops team. A stranger cannot audit our ops team. They can read a public chain.

“You’re only storing a hash, that’s nothing.”

That’s the point. Because the seal is tiny, we can stamp every check when it happens — not later, when the lawsuit starts, when a backdated seal is worthless. Cheap is what lets us be honest all the time instead of honest when it’s convenient.

“Why not your own chain / Hyperledger / a server in Dhaka?”

A chain we run is still a notebook with our pen, just with extra ceremony. The whole reason we’re doing this is that *we* are the party whose word is in doubt. Settlement on Ethereum — we write cheaply on Base, which dumps a summary down to Ethereum — means undoing a stamp means fighting Ethereum, not phoning our intern.

04 The wax seal

You take the whole answer — what we found, what we couldn’t check, which rules we used — and you freeze the wording so it comes out the same every time. Then you run it through a blender that always spits out the same tiny seal if the letter is the same, and a totally different seal if anyone so much as moved a comma. That’s a hash. You cannot read the letter from the seal. You cannot make a second letter that accidentally gets the same seal. Anyone who later has the letter can blend it themselves and see if the wax matches the board.

That’s also why the AI isn’t allowed to write an essay. Essays fidget. Seals need a letter that sits still.

05 The tournament bracket (why one stamp covers a thousand checks)

Pinning every seal one-by-one would get expensive and fiddly. So we do a tournament. Each check is a player. Every round, two seals combine into a parent seal, all the way up to one champion seal. We pin the champion. Later, to prove your match was in that tournament, you only show the path of scores up to the final — a handful of siblings, not the other 999 letters. Strangers can replay that path against the champion that’s already on the board. They never need our filing cabinet.

People who work on this call it a Merkle tree. Same shape as the log that proves which HTTPS certificates exist. You don’t have to say RFC numbers unless someone is showing off. You do have to say: one public stamp, many private letters, and a short proof that your letter was in the batch.

06 We used the public form, not a homemade one

There’s already a standard on Base for “someone attested this blob at this time.” It’s called EAS — Ethereum Attestation Service. Think of it as a public notary form everyone already knows how to read. We didn’t invent our own weird “store 32 bytes” contract, because then a stranger would have to trust our homework. With EAS they can use the public explorer like anyone else.

We only wrap it to say who is allowed to stamp. Membership can be paused. Old stamps stay. You don't get to rewrite September by firing someone in November.

07 Two signatures, and a receipt you can't sell

A stamp we make alone proves we *said* it. It doesn't prove the other party agreed. So the other party can sign the same letter in their wallet, looking at a readable summary — not a blob of hex a phishing site could swap. Their key never comes to us. Two signatures beat a diary entry.

We can also mint a receipt that means "this named company ran this check." It is not a coin. It cannot be transferred. If you can buy someone else's homework, it stopped being homework. If a judge asks where the token is, you say: there isn't one, on purpose.

08 Why Base, without a TED talk

Ethereum itself is expensive if you write on it all day. Base is the cheap notebook that still hands the final copy to Ethereum. If Base tries to publish a lie, anyone can challenge it on Ethereum during the window. After that, changing a stamp means attacking Ethereum, not guessing our admin password. That's the whole "why not a private chain" answer, dressed in one extra sentence.

We sign the letter with one kind of key in our software, and we pay for the stamp with a different Ethereum key in a vault. Steal one, you don't get the other. Old stamps don't die if we rotate a key. New contract versions get new addresses — nobody silently swaps the rules under an old seal.

09 How a stranger checks without logging into us

1. They get the letter (from the customer, a lawyer, an export — not from our login wall).
2. They blend it themselves. Same blender, same frozen wording.
3. They look at the noticeboard. Is that seal there, at that time?
4. If it was part of a batch, they check the little path up to the champion seal.
5. If the other party co-signed, they check that signature too.
6. If they want to argue about truth, not tampering, they can re-run the AI. That's a different fight.

If the check requires our website, it isn't independent. Don't build a story that needs us in the room.

10 Say it in this order

1. The defence is a date. We're paid to hold the date. That's a conflict.
2. Postgres is our pen. The chain is a noticeboard we don't own.
3. Append-only is still our password. A private chain is the same trick.
4. We pin a wax seal, not the picture, so we can stamp every check, not just the lawsuits.

5. Tournament of seals, one public champion, short proof you were in the draw.
6. Public notary form (EAS), not a homemade contract. Optional second signature. No coin.
7. Anyone can check without logging into us.

IF YOU REMEMBER ONE LINE

We need a witness with no privileged writer. That’s the one thing a database cannot be.